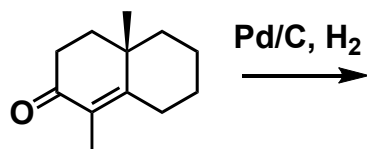
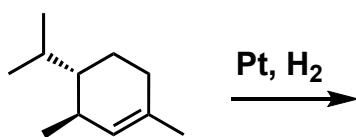
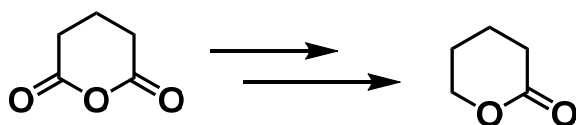


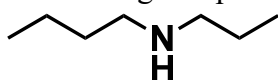
1. Give the major products for the following reactions and explain the stereochemistry:



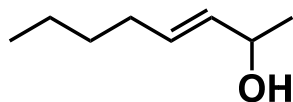
2. Using necessary inorganic or organic reagents, complete the following transformation.



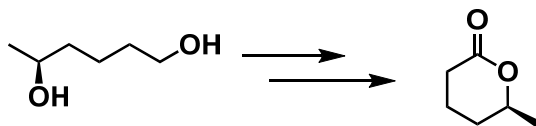
3. Using organic reagents that contain three carbon atoms or less and necessary inorganic reagents, synthesize the following compound:



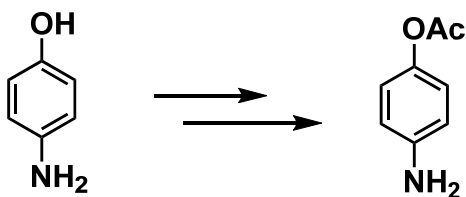
4. Using organic reagents that contain three carbon atoms or less and necessary inorganic reagents, synthesize the following compound:



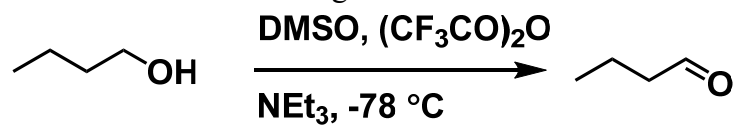
5. Using necessary inorganic or organic reagents, complete the following transformation.



6. Using necessary inorganic or organic reagents, complete the following transformation.



7. Draw the mechanism for the following reaction:



8. Using salicylic acid and organic compounds containing less than 4 carbon atoms (3C and less) along with necessary inorganic reagents, suggest an efficient synthetic route for the following molecule.

